



Mustafa Sibai

Principal Software Engineer

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About

Full Stack Developer, Games Lecturer, App, Game, and Engine Developer with a total of 14 years of experience.

Shipped multiple mobile apps, and game titles on a variety of platforms. Wrote multiple game engines and softwares in C++, C#, JavaScript, TypeScript, Python, and more.

Experience

Cheif Tecnology Officer

Feb 22 - Present

Couch Heroes

- Led and mentored a team of eight people to create an MMO game in Unreal Engine 5.
- Defined the organization's technology stack.
- Designed the networking server architecture to support thousands of players.
- Designed and developed 30+ Microservices in Typescript, Node.js, Express, MongoDB, Redis, and WebSocket.
- Designed and implemented the architecture for various servers, including matchmaking, marketplace, social, chat, and inventory management systems. Developed and maintained these services using TypeScript, Node.js, Express, MongoDB, WebSockets, and Unreal Engine 5.
- Designed and developed the organization's CI/CD pipelines in Microsoft Azure for building, testing, and development.
- Designed and implemented the architecture for multiple web applications, developing and maintaining them using React, Redux, Redux Saga, and Tailwind CSS.
- Designed and implemented the architecture for our game launcher, developing and maintaining it with TypeScript, Node.js, Express, MongoDB, WebSockets, and Unreal Engine 5.
- Create a private DNS and mapped the organization's servers to domains using Nginx reverse proxy.
- Monitored the organization's servers using Prometheus, Grafana, and Uptime Kuma and created Slack and Email alerts.
- Gathered the organization's requirements and created a three-year roadmap, including quarterly plans for the dev team to follow.
- Created and maintained Onboarding, Dev, DevOps, and Security handbooks for the organization.
- Created and maintained documentation for the dev team to follow, which included the requirements of the organization's products and their architecture.
- Secured the organization's servers behind a self-hosted VPN through WireGuard.
- Containerized the organization's services using Docker and scaled them both vertically and horizontally using Kubernetes.
- Created and managed the organization's infrastructure and services through Microsoft Azure.
- Conducted one-on-one code reviews with both junior and senior engineers and wrote performance reviews for the dev team.
- Conducted various workshops on Git, Docker, Kubernetes, and CI/CD pipelines.
- Solved various Git issues for a team of 40 people with diverse backgrounds, including game design, art, programming, audio, and QA.

Education

Essex University Online
MSc Computer Science
2022 - 2024

Emirates Aviation University
HND Computer Science
2010 - 2012

Emirates Aviation University
NC Aerospace Engineering
2009 - 2010

Technical Skills

Game Development	12 yrs
Frontend Development	8 yrs
Backend Development	8 yrs
Engine Development	4 yrs
Native Mobile Development	1 yrs
OS Development	1 yrs

C#	12 yrs
C++	8 yrs
JavaScript	8 yrs
TypeScript	8 yrs
Python	4 yrs
Shell Script	4 yrs
HTML / CSS	4 yrs
Java	2 yrs
Swift	1 yrs
Go	1 yrs
Rust	1 yrs
x86 Assembly	1 yrs

React / React Native	6 yrs
Redux / Redux-Saga	4 yrs
Gatsby	2 yrs
Vue.js	1 yrs
Flutter	1 yrs

Node.js / Express.js	8 yrs
REST API	8 yrs
MongoDB	6 yrs
SQL	4 yrs
GraphQL	2 yrs
Django / Flask	1 yrs

Unity Engine	10 yrs
Unreal Engine	3 yrs
Godot	1 yrs

DirectX 11	3 yrs
OpenGL 4.5	3 yrs
Vulkan	1 yrs

Games Lecturer

Jun 17 - Present

SAE Institute Dubai

- Taught the Bachelor's program in video game programming, including the following topics.
- Unity Engine with C#, Unreal Engine 5 with C++ and Blueprints.
- AI, including State Machine, A* Pathfinding, Behaviour Trees, GOAP, and Vision Cones.
- Multiplayer using TCP protocol in C# Sockets.
- 3D game math, including Vectors and Matrices.
- Computer Graphics, Raytracing, HLSL, GLSL, and CG Shaders.
- Project management using Agile methodology.

Lead Game Developer

Feb 22 - Sep 22

Couch Heroes

- Led a team of three game developers and managed the project by implementing Agile development methodology.
- Designed and developed multiple microservices in Typescript, Node.js, Express, MongoDB, Redis, and WebSocket.
- Designed and developed an authentication microservice using JWT and OAuth 2.0 flow.
- Designed and developed a player registration microservice to allow players to register in our game.
- Designed and developed a social microservice that allowed players to add friends, remove friends, accept friend requests, reject friend requests, and block friends.
- Designed and developed a player account management microservice to allow players to update their information such as email, password, profile picture, etc...
- Designed and developed an inventory system to keep track of player items in the game.
- Designed and developed a frontend dashboard for players to register, sign in, update their information, and more.
- Designed and developed a marketplace microservice to allow players to buy, sell, and trade items in the game.
- Designed and developed a RESTful API that uses the microservices architecture to allow partners to integrate our API into their games.
- Maintained and managed 40+ Git repositories and created different environments for our services, such as Nightly, Dev, Stage, and Prod.

Senior React Native Developer

Jun 21 - Jan 22

Alshaya Group

- Optimize the performance of Victoria Secret app.

React Native Developer

Sep 20 - Nov 20

Emaar Entertainment LLC

- Optimized the App by reducing asset sizes and converting them to SVG format.
- Added app localisation and support for RTL (Right-to-Left) languages and translated all text into Arabic.

WinSock 2	8 yrs
WebSocket	6 yrs
Mirror / PUN	4 yrs
Microsoft Azure	7 yrs
Google Cloud	5 yrs
Firebase	2 yrs
AWS	1 yrs
Git	8 yrs
DevOps CI/CD	8 yrs
Docker	6 yrs
Kubernetes	6 yrs
Scrum Master	4 yrs
Windows	8 yrs
MacOS	8 yrs
Android	8 yrs
Linux	8 yrs
IOS	4 yrs

Founder - CTO

Oct 19 - Feb 22

Blue Monocle Software

- Started a game, animation, and App development studio.
- Developed and produced Apps and games in C++, C#, Unity, JavaScript, React, React Native, Node.js, and Gatsby.
- Recruited and managed programmers, game designers, artists, and audio engineers.

Founding Partner - CTO

Oct 19 - Jan 21

Beautify

- Defined the company technology and hired a team to create a mobile-based booking and E-commerce platform on both Android and IOS.
- Managed the entire app development cycle from start to finish while also helping develop the application with the programming and designing team.

CTO

Apr 19 - Jul 19

Coded Minds

- Defined the company technology and hired a team to create the web-based / Android and IOS education platform.
- Recruited and led a team in designing a STEM-based curriculum and created VR/AR-based tools and educational games.

Lead Game Developer

Feb 18 - Aug 18

SRH Studios

- Recruited and led a team of game programmers and artists to develop mobile games, VR and AR content using the Unity game engine for both IOS and Android.

Network Engineer

Mar 17 - Jun 17

Digital Tree Software

- Designed and implemented server/client-based network code using the Unity game engine.

Game Developer

Sep 12 - Oct 17

Vault 16 Software

- Designed, implemented, and published multiple games for various clients in C++ and C# on Android and IOS.

Commercial Projects

Couch Heroes

Feb 22 - Present

Unreal Engine 5 | [Windows](#)

- An unannounced MMO game built in Unreal Engine 5.

Victoria Secret

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

H&M

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

FootLocker

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

American Eagle

Jun 21 - Jan 22

TypeScript | React Native • [Apple Store](#) | [Google Play Store](#)

Bugged Dungeon

Jan 21 - Feb 22

C# | Unity • [Steam](#) | [Apple Store](#) | [Google Play Store](#)

- 3D puzzle dungeon game.

Reel Cinema App

Sep 20 - Nov 20

JavaScript | React Native • [Apple Store](#) | [Google Play Store](#)

- Movie booking app.

BlackSky

Jul 20 - April 22

C# | Unity • [Steam](#) | [Apple Store](#) | [Google Play Store](#)

- 3D space match match action game.

River Run

Jul 18 - Aug 18

C# | Unity • [Apple Store](#) | [Google Play Store](#)

- 3D endless boat runner game.

Jump AR

Mar 18 - May 18

C# | Unity • [Apple Store](#) | [Google Play Store](#)

- 3D AR game with jump mechanics.

Race Start 2

Feb 15 - Aug 15

C# | Unity | Photon Networking • [Apple Store](#) | [Google Play Store](#)

- A 2D singleplayer/multiplayer platformer game.

Personal Engines

Blue Flame Engine

Mar 16 - Mar 18

C++ | OpenGL 4.5 | DirectX 11 | Windows | Linux | Android | Web • [GitHub](#)

- Developed an Entity/Component-based game engine that supports 2D and 3D rendering, Forward rendering, sprite batch rendering, Post-processing, Phong shading, point, spot, and directional light system, 3D math library, 2D physics engine, A* pathfinding, networking, and a C++ scripting system.

Vault Engine

Mar 14 - Apr 15

C# | XNA | Windows • [GitHub](#)

- Developed a 2D tile-based game engine which supports frustum culling, multithreaded A* pathfinding using a custom job system, and networking with server/client-based architecture using TCP and UDP protocol written in Winsocket2.

Game Jam Projects

Bugged Dungeon

Jan 21

C# | Unity • [Global Game Jam](#)

- Developed a Lost & Found themed game with a team of two.

D-I-Why? Armour Guy

Jan 20

C# | Unity • [Global Game Jam](#)

- Developed a repair themed game with a team of nine.

Pounce Hug!

Jan 19

C# | Unity • [Global Game Jam](#)

- Developed a home themed game with a team of six.

Neon Fever: The Game

Jan 18

C# | Unity • [Global Game Jam](#)

- Developed a transmission themed game with a team of six.

Caved In

Jan 17

C# | Unity • [Global Game Jam](#)